MSLAN: A Two-Branch Multidirectional Spectral–Spatial LSTM Attention Network for Hyperspectral Image Classification

Tiecheng Song, Yuanlin Wang, Chenqiang Gao, Haonan Chen, *Senior Member, IEEE*, and Jun Li, *Fellow, IEEE*

***Abstract*— Recurrent neural networks (RNNs) have been widely used for hyperspectral image (HSI) classification via sequence modeling. However, most of the RNN methods focus on modeling long-range dependencies along the spectral direction without fully exploring multidirectional dependencies in the joint spectral–spatial domain. To tackle this issue, we propose MSLAN, a two-branch multidirectional spectral–spatial long short-term memory (LSTM) attention network, for HSI classification. In particular, we employ LSTMs to extract six-directional spatial–spectral features that simultaneously capture the spectral–spatial dependencies along with different directions. We then design an attention-based feature fuse module to integrate these directional features, followed by a fully connected layer with cross-entropy loss for classification. In addition, we incorporate an auxiliary branch into our model to enhance the generalization capability. In this branch, random spatial shuffle and a cosine loss are explored for feature consistency learning by taking into account the varying spatial distributions. The resulting two branch networks, sharing the same network structure and weights, are incorporated into a unified deep learning architecture for training. Experiments show the superiority of MSLAN to the state-of-the-art methods for HSI classification with limited training samples.**

***Index Terms*— Attention, deep learning, hyperspectral image (HSI) classification, long short-term memory (LSTM), recurrent neural network (RNN), spectral–spatial feature.**

I. Introduction

HYPERSPECTRAL images (HSIs), which contain hundreds of narrow and contiguous spectral bands, provide rich information to identify a variety of materials and have been widely used in precision agriculture, geological exploration, environmental monitoring, and so on. Among them, HSI classification, aiming to assign a land-cover label to each pixel, is one of the most vibrant yet challenging research fields in the hyperspectral community [1]. The main challenges include the large spatial variability of spectral signatures, the limited number of available training samples, and the high dimensionality of hyperspectral data [2].

A number of HSI classification methods have been proposed in the literature. One straightforward method is to feed the original spectral signatures into advanced classifiers, such as support vector machine and random forest [3]. However, it can easily lead to the curse of dimensionality (i.e., the Hughes phenomenon) due to the unbalance between limited training samples and high-dimensional features. To alleviate this problem, dimensionality reduction techniques, including feature selection [4] and feature extraction [5], were developed. The latter includes the principal component analysis (PCA), the linear discriminant analysis, and manifold learning [6], [7]. As spectral signatures often exhibit interclass similarity and intraclass dissimilarity, spatial context information has been leveraged to improve classification performance [8]. In this respect, local binary patterns [9], Gabor filters [10], and morphological profiles [11] were typically used to extract spatial features. Markov random fields [12], edge-preserving filtering [13], and superpixel segmentation [14] were explored as spatial constraints to optimize or fuse the classification results.

Nevertheless, the aforementioned methods use handcrafted features that require domain knowledge and expert experience, limiting their applications in other different scenarios. Also, the extracted features are shallow, and they are difficult to well represent HSIs that lie in a nonlinear and complex feature space [15]. To overcome these issues, deep learning-based methods have been developed to extract hierarchical spectral or spectral–spatial features [16]. For example, Chen *et al.* applied stacked autoencoders [17] and deep belief networks [18] to learn spectral–spatial features followed by a logistic regression layer for HSI classification. Hu *et al.* [19] used 1-D convolutional neural networks (CNNs) to classify HSIs directly in the spectral domain. Yue *et al.* [20] incorporated 2-D CNNs into a joint spectral–spatial classification framework. Chen *et al.* [21] employed 3-D convolution kernels to extract spectral–spatial features from HSI cubes. Roy *et al.* [22] designed a hybrid spectral CNN (HybridSN) by assembling spectral–spatial 3-D CNNs and spatial 2-D

Manuscript received February 13, 2022; revised April 30, 2022; accepted May 13, 2022. Date of publication May 18, 2022; date of current version May 31, 2022. This work was supported in part by the National Natural Science Foundation of China under Grant 61702065 and in part by the Chongqing Research Program of Basic Research and Frontier Technology under Grant cstc2018jcyjAX0033. The work of Haonan Chen was supported by Colorado State University. *(Corresponding author: Haonan Chen.)*

Tiecheng Song, Yuanlin Wang, and Chenqiang Gao are with the School of Communication and Information Engineering and the Chongqing Key Laboratory of Signal and Information Processing, Chongqing University of Posts and Telecommunications, Chongqing 400065, China.

Haonan Chen is with the Department of Electrical and Computer Engineering, Colorado State University, Fort Collins, CO 80523 USA (e-mail: haonan.chen@colostate.edu).

Jun Li is with the School of Computer Science and the Hubei Key Laboratory of Intelligent Geo-Information Processing, China University of Geosciences, Wuhan 430078, China.

Digital Object Identifier 10.1109/TGRS.2022.3176216

1558-0644 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:14:31 UTC from IEEE Xplore. Restrictions apply.

CNNs. Zhong *et al.* [23] proposed a spectral–spatial residual network (SSRN) using two consecutive spectral and spatial residual blocks. Li *et al.* [24] built a CNN model based on adjacent pixel pairs. Zhang *et al.* [25] constructed a diverse region-based CNN to learn spectral–spatial features. Hao *et al.* [26] presented a two-stream deep architecture with a class-specific fusion scheme: one stream using the stacked denoising autoencoder to handle spectral pixels and the other stream using CNNs to process image patches. Besides, covariance matrix representations [27], graph convolutional networks [28], [29], and generative adversarial networks [30] were also used for HSI classification.

The spectral–spatial features extracted from HSIs corresponding to different spectral bands and spatial positions have different importance. For this sake, attention mechanisms [31], [32] have been recently introduced for HSI classification. Haut *et al.* [33] introduced attention to a residual network by considering both bottom-up and top-down visual factors. Zhu *et al.* [34] embedded a sequential spectral–spatial attention module into a residual block to learn informative features. Sun *et al.* [35] incorporated the self-attention mechanism into a deep network model, which was composed of two sequential spectral and spatial modules. Hang *et al.* [36] proposed a two-branch spectral–spatial attention network with each branch focusing on discriminative channels or spatial positions. Xi *et al.* [37] proposed a multidirection network where 3-D squeeze-and-excitation (SE) residual blocks were devised in each stream to consecutively learn spectral and spatial features. Meanwhile, the attentional spectral prior was embedded by considering spectral correlations. Roy *et al.* [38] presented an attention-based adaptive spectral–spatial kernel improved residual network (A^2S^2K-ResNet) by learning selective 3-D convolutional kernels and adopting an efficient feature recalibration mechanism. In addition, Roy *et al.* [39] proposed a fusion of SE network (FuSENet) using global average and max pooling to generate attention vectors. Xue *et al.* [40] designed an attention-based second-order pooling operator to learn representative high-order features.

Recently, the recurrent neural network (RNN) and its variants, including long short-term memory (LSTM) and the gated recurrent unit (GRU), have been used to process hyperspectral data.[1] Different from convolutional filters used in CNNs which only capture local dependencies, RNNs treat hyperspectral pixels as sequential data and model long-range dependencies with recurrent connections. For the first time, Mou *et al.* [41] built an RNN model to process hyperspectral pixels. Wu and Prasad [42] proposed a convolutional RNN that was composed of several convolutional and recurrent layers to learn spectral features. Hang *et al.* [15] proposed a cascaded RNN (CasRNN) model with GRUs to explore the redundant and complementary information of hyperspectral pixels. Paoletti *et al.* [43] recently investigated the scalable implementation of a novel RNN model based on simple recurrent units. To incorporate spatial information, Hang *et al.* [15] extended CasRNN to a spectral–spatial version by adding some convolutional layers before RNN layers. Convolutional LSTM (ConvLSTM) [44], an extension of LSTM with convolutional structures in input-to-state and state-to-state transitions, was explored in [45] by considering both spectral and spatial features. RNNs with 3-D convolutions were also utilized in [46] and [47] to capture rich context information. Besides, Xu *et al.* [48] proposed a spectral–spatial unified network (SSUN) where spectral and spatial features were learned via a band grouping-based LSTM model and a multiscale CNN model, respectively.

Although the above convolution-based RNNs have considered the spectral–spatial features, they focus on modeling long-range dependencies along the spectral direction using the recurrent connections and do not explicitly model such dependencies along with different spatial directions. The directional changes of land-cover distributions tend to make these methods wrongly classify pixels in heterogeneous regions, especially at the edge of cross-class regions where the relative pixel positions of two land-cover classes can be changed from left to right or from up to down and so on. Few attempts were made to capture spatial sequence dependencies using RNNs for HSI classification. For example, Zhang *et al.* [49] constructed an RNN model based on local and nonlocal spatial sequence features. In this model, the sequence features are sorted before being fed into the RNN model, and hence, the directional information is lost. Shi and Pun [50] explored spatial RNNs [51], [52] to learn spatial dependencies of nonadjacent image patches along with multiple directions at different scales. Similarly, Liu *et al.* [53] designed an RNN to model spatial dependencies between different neighborhood pixels along the row direction. However, spectral dependencies are not explored in both methods. Zhou *et al.* [54] fed the rows of image patches (derived from the first principle component of HSIs) into one LSTM to learn spatial features and fed spectral vectors into the other LSTM to learn spectral features. Hence, this method only captures spatial dependencies along the column direction. Recently, Wang *et al.* [55] employed the band grouping-based LSTM and spatial grouping-based ConvLSTM to extract spectral and spatial features, respectively. However, they only considered unidirectional dependency relationships.

Inspired by the above works, in this article, we propose a novel LSTM-based attention network to fully explore multidirectional dependencies in the joint spectral–spatial domain. The proposed network model shows superior performance over the state-of-the-art methods for HSI classification when limited training samples are available. The contributions of this article are summarized as follows.

1) A two-branch multidirectional spectral–spatial LSTM attention network (MSLAN) is proposed for HSI classification. MSLAN is an end-to-end deep neural network model that incorporates spectral–spatial feature extraction, feature fusion, and classifier learning into a unified framework for training.
2) Multidirectional dependencies in the joint spectral–spatial domain are modeled using spectral and spatial LSTMs. An attention module is designed to fuse directional spectral–spatial features. In addition, parameter

[1] RNN, LSTM, and GRU are sometimes collectively called RNN hereinafter.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:14:31 UTC from IEEE Xplore. Restrictions apply.

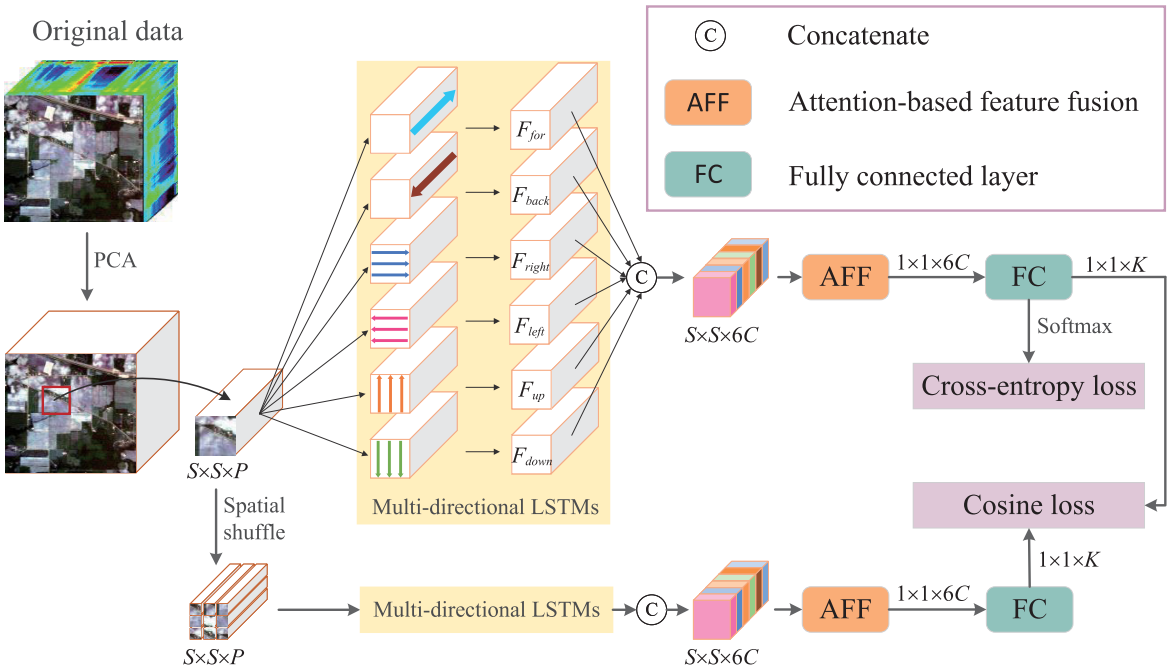


Fig. 1. Flowchart of the proposed MSLAN for HSI classification. The spatial-shuffle branch is only used in the training processing. The weights of multidirectional LSTMs, AFF, and FC in both branches are shared. The feature size is marked in the figure, and K is the number of land cover classes.

sharing is adopted in spatial LSTMs to reduce the number of model parameters and improve training efficiency.

3) An auxiliary branch based on random spatial shuffle and a cosine loss is incorporated into our model for feature consistency learning by taking into account the varying spatial distributions. The two branch networks share the same network structure and weights, making MSLAN a lightweight model and generalize well for HSI classification with limited training samples.

The rest of this article is organized as follows. The proposed model MSLAN is introduced in Section II. The classification experiments are presented in Section III. Finally, the conclusions are drawn in Section IV.

II. Architecture of MSLAN for HSI Classification

Fig. 1 shows the flowchart of MSLAN for HSI classification. First, we perform PCA to reduce the dimensionality of hyperspectral data. Then, we crop the input HSI into 3-D cubes and represent each hyperspectral pixel by its surrounding HSI cube denoted as $F \in \mathbb{R}^{S\times S\times P}$, where $S \times S$ is the spatial size of HSI cubes and P is the number of principal components. Then, we pass these cubes to the MSLAN model to learn multidirectional spectral–spatial features for HSI classification. There are two branches in MSLAN. In the first branch, the input cube passes through a multidirectional LSTM-based module and an attention-based feature fusion (AFF) module, followed by a fully connected layer with the cross-entropy loss for classification. The second branch has the same network structure and weights as those in the first branch except that the input cube is initially spatially shuffled and a cosine loss is used for feature consistency learning. In the following, we first introduce the regular LSTM and then detail the proposed network model.

A. LSTM

RNNs are a type of neural network that uses cyclical connections to learn context dependencies of sequential data. Basic RNNs are not good at modeling long-term dependencies due to the vanishing and exploding gradients. To deal with these problems, LSTM [56] replaces the recurrent hidden node in RNNs with a "memory block" and has shown promise performance in many applications [41], [52]. Fig. 2 shows the structure of LSTM, which contains four important parts: an input gate, a forget gate, an output gate, and a self-recurrent connection. The memory cell stores information on current and previous time steps and three gates control the information flow in the network. The calculation process in the LSTM model at time step t is formulated as follows:

$$\boldsymbol{i}_t = \sigma(\boldsymbol{W}_{xi}\boldsymbol{x}_t + \boldsymbol{W}_{hi}\boldsymbol{h}_{t-1} + \boldsymbol{b}_i) \tag{1}$$
$$\boldsymbol{f}_t = \sigma(\boldsymbol{W}_{xf}\boldsymbol{x}_t + \boldsymbol{W}_{hf}\boldsymbol{h}_{t-1} + \boldsymbol{b}_f) \tag{2}$$
$$\boldsymbol{o}_t = \sigma(\boldsymbol{W}_{xo}\boldsymbol{x}_t + \boldsymbol{W}_{ho}\boldsymbol{h}_{t-1} + \boldsymbol{b}_o) \tag{3}$$
$$\boldsymbol{g}_t = \tanh(\boldsymbol{W}_{xg}\boldsymbol{x}_t + \boldsymbol{W}_{hg}\boldsymbol{h}_{t-1} + \boldsymbol{b}_g) \tag{4}$$
$$\boldsymbol{c}_t = \boldsymbol{f}_t \odot \boldsymbol{c}_{t-1} + \boldsymbol{i}_t \odot \boldsymbol{g}_t \tag{5}$$
$$\boldsymbol{h}_t = \boldsymbol{o}_t \odot \tanh(\boldsymbol{c}_t). \tag{6}$$

Here, $\boldsymbol{W}_{xi}$, $\boldsymbol{W}_{hi}$, $\boldsymbol{W}_{xf}$, $\boldsymbol{W}_{hf}$, $\boldsymbol{W}_{xo}$, $\boldsymbol{W}_{ho}$, $\boldsymbol{W}_{xg}$, and $\boldsymbol{W}_{hg}$ are weight matrices. $\boldsymbol{b}_i$, $\boldsymbol{b}_f$, $\boldsymbol{b}_o$, and $\boldsymbol{b}_g$ are bias vectors. $\boldsymbol{x}_t$, $\boldsymbol{i}_t$, $\boldsymbol{f}_t$, $\boldsymbol{o}_t$, $\boldsymbol{g}_t$, $\boldsymbol{c}_t$, and $\boldsymbol{h}_t$ are the input, the input gate, the forget gate, the output gate, the candidate memory cell, the memory cell, and the hidden state at time step t, respectively. The symbol $\odot$ denotes the elementwise multiplication. $\sigma(x) = 1/(1+e^{-x})$ is the sigmoid function, and $\tanh(x) = (e^x - e^{-x})/(e^x + e^{-x})$ is the hyperbolic tangent function. Since the same model parameters are used at different time steps, the number of parameters in RNNs does not grow with the number of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:14:31 UTC from IEEE Xplore. Restrictions apply.

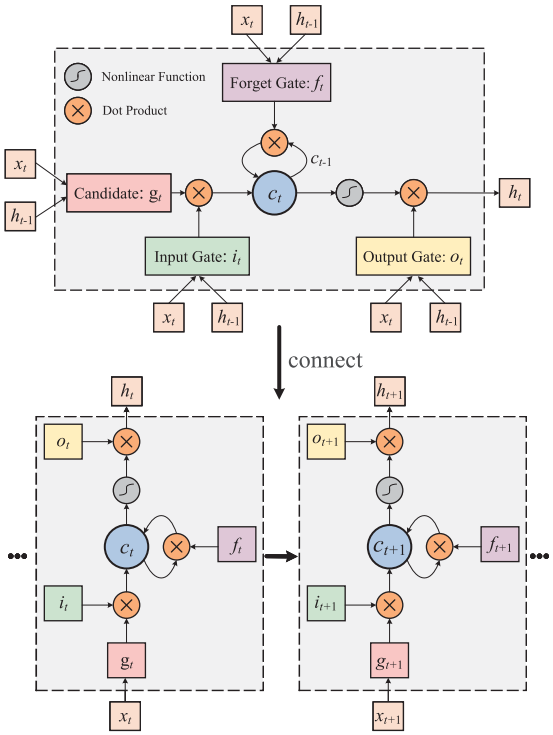


Fig. 2. Structure of an LSTM network model.

time steps. RNNs can be optimized by the backpropagation through time (BPTT) algorithm [56].

As shown in Fig. 2, each LSTM unit corresponds to a time step, and multiple LSTM units are connected in series to constitute an LSTM-based network. For each LSTM unit, its output is affected by the hidden state and the cell state of previous LSTM units. In this way, LSTMs can discover long-term dependencies and potential patterns in sequential data.

B. Multidirectional LSTM-Based Module

Conventional CNNs perform convolution and pooling operations on local image regions without explicitly modeling long-range dependencies among nonadjacent pixels or bands. However, such dependencies are important to describe HSIs, which often have complex spatial and spectral variations. Although several RNN-based methods have been developed toward this goal [50], [53]–[55], the multidirectional dependencies in the joint spectral–spatial domain are not fully explored, and their performance is not satisfactory. To bridge this gap, we build a multidirectional LSTM-based module to extract spectral–spatial features and capture their dependencies in six directions, i.e., forward, backward, rightward, leftward, upward, and downward. Note that, in the computer vision field, Pan *et al.* [57] proposed a spatial CNN to achieve effective information propagation between pixels across rows and columns in a layer. Liu *et al.* [58] proposed a spatial propagation network with four directions to learn the affinity matrix. These two methods and ours have some similar motivations, but they only consider multidirectional information propagation in the spatial dimension. Also, the network structures and the addressed tasks are different.

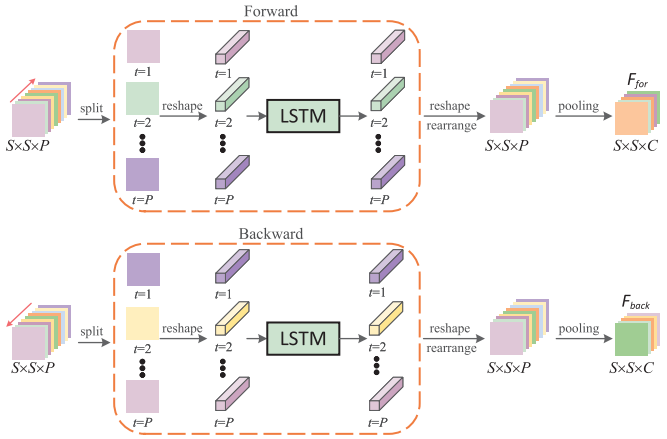


Fig. 3. Spectral LSTMs along forward and backward directions.

Specifically, we use two spectral LSTMs to capture the spectral dependencies along with the forward and backward directions, respectively. As shown in Fig. 3, we first split the input cube $F \in \mathbb{R}^{S\times S\times P}$ into P feature maps denoted as $M_i \in \mathbb{R}^{S\times S\times 1}$ $(i = 1, \ldots, P)$, each corresponding to a time step. Then, we reshape each feature map into a vector and feed the sequence of P vectors into an LSTM whose output has the same dimensions as its input. After reshaping the output vectors and arranging them into a cube of size $S \times S \times P$, we perform 3-D adaptive averaging pooling[2] on this cube to yield the "forward" feature $F_{\text{for}} \in \mathbb{R}^{S\times S\times C}$. Different from conventional convolutions, this pooling requires no additional parameters to be learned. Similarly, we can reverse the order of feature maps and feed the resulting sequence into another LSTM to learn the "backward" feature $F_{\text{back}} \in \mathbb{R}^{S\times S\times C}$.

Furthermore, we use four spatial LSTMs to capture the spatial dependencies along rightward, leftward, upward, and downward directions, respectively. Because the four LSTMs work similarly except for the input sequences, we only take the "rightward" LSTM as an example to detail the process. As shown in Fig. 4, we first split the input cube $F \in \mathbb{R}^{S\times S\times P}$ into $S \times S$ hyperspectral pixels denoted as $R_j \in \mathbb{R}^{1\times 1\times P}$ $(j = 1, \ldots, S \times S)$. Then, we scan each row of pixels from left to right as a sequence (each pixel corresponds to a time step) and feed it into an LSTM. Here, each row of pixels goes through an LSTM, and a total of S LSTMs share the same weights to improve training efficiency. This sharing scheme also alleviates overfitting when only limited training samples are available. Finally, we rearrange the outputs of all LSTMs (in order of their positions in the input cube) as the "rightward" feature $F_{\text{right}} \in \mathbb{R}^{S\times S\times C}$. By changing the scanning order or direction, we can, in a similar way, obtain the other three features, i.e., F_{left}, F_{up}, and F_{down} for leftward, upward, and downward directions, respectively.

Finally, we concatenate all the directional features as the output of the multidirectional LSTM-based module, i.e.,

$$F_{\text{joint}} = \text{ccat}(F_{\text{for}}, F_{\text{back}}, F_{\text{right}}, F_{\text{left}}, F_{\text{up}}, F_{\text{down}}) \tag{7}$$

[2]We use the built-in function AdaptiveAvgPool3d in PyTorch where the values such as kernel size and stride for averaging pooling are automatically determined by the sizes of input and output.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:14:31 UTC from IEEE Xplore. Restrictions apply.

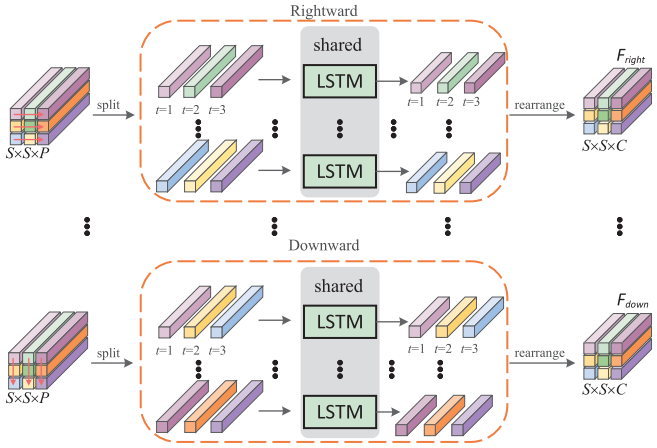


Fig. 4. Spatial LSTMs along rightward, leftward, upward, and downward directions ($S = 3$ is shown only for illustrative purposes).

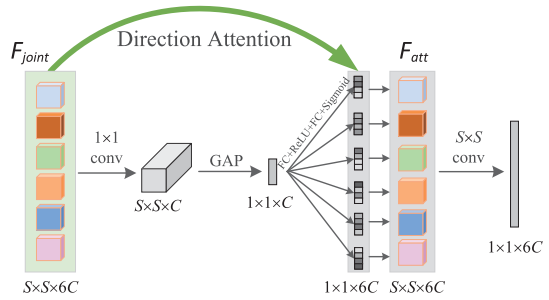


Fig. 5. Structure of the AFF module.

where ccat denotes the concatenation operation. Therefore, $F_{\text{joint}} \in \mathbb{R}^{S\times S\times 6C}$ contains the joint six-directional spectral–spatial dependencies for hyperspectral pixels in an HSI.

C. Attention-Based Feature Fusion Module

Different directional information has different effects on classification performance. To effectively select and integrate these directional features in (7), we design the AFF module. As shown in Fig. 5, different from the SE block in [32], AFF embeds a multibranch network that encourages each branch to learn attention weights for one specific direction (i.e., AFF learns groupwise direction attention).

Specifically, given the input F_{joint}, we adopt 1×1 convolutions to generate the feature cube $F_{\text{fuse}} \in \mathbb{R}^{S\times S\times C}$. Then, we perform global average pooling (GAP) to obtain channelwise statistics $\boldsymbol{v} \in \mathbb{R}^{C}$. Next, we feed $\boldsymbol{v}$ into a multibranch feedforward neural network to generate six direction-attention vectors $\boldsymbol{z}_{\text{dir}} \in \mathbb{R}^{C}$, where the subscript "dir" indicates one of the six directions. The above process can be formulated as

$$\boldsymbol{v}'_{\text{dir}} = \text{ReLU}(\text{FC}(\boldsymbol{v})) \tag{8}$$

$$\boldsymbol{z}_{\text{dir}} = \text{Sigmoid}\big(\text{FC}\big(\boldsymbol{v}'_{\text{dir}}\big)\big) \tag{9}$$

where ReLU$(\cdot)$ and Sigmoid$(\cdot)$ are two activation functions and FC is a fully connected layer. After that, we use the attention vector $\boldsymbol{z}_{\text{dir}}$ to rescale the directional feature $F_{\text{dir}} \in \mathbb{R}^{S\times S\times C}$. To be specific, let $z^c_{\text{dir}} \in \mathbb{R}$ be the cth element of $\boldsymbol{z}_{\text{dir}}$ and $F^c_{\text{dir}} \in \mathbb{R}^{S\times S\times 1}$ be the cth feature map of F_{dir} $(c = 1, \ldots, C)$. For each direction dir, we perform channelwise multiplication between the feature map F^c_{dir} and the scalar z^c_{dir}, i.e.,

$$\tilde{F}^c_{\text{dir}} = z^c_{\text{dir}} \cdot F^c_{\text{dir}} \tag{10}$$

and concatenate the results of C channels as

$$F^{\text{att}}_{\text{dir}} = \text{ccat}\big(\tilde{F}^1_{\text{dir}}, \tilde{F}^2_{\text{dir}}, \ldots, \tilde{F}^C_{\text{dir}}\big) \tag{11}$$

where $F^{\text{att}}_{\text{dir}} \in \mathbb{R}^{S\times S\times C}$. After processing all six directions using attention, we obtain the concatenated feature by

$$F_{\text{att}} = \text{ccat}\big(F^{\text{att}}_{\text{for}}, F^{\text{att}}_{\text{back}}, F^{\text{att}}_{\text{right}}, F^{\text{att}}_{\text{left}}, F^{\text{att}}_{\text{up}}, F^{\text{att}}_{\text{down}}\big) \tag{12}$$

where $F_{\text{att}} \in \mathbb{R}^{S\times S\times 6C}$. Finally, we apply a convolutional layer (group convolution) to the concatenated feature F_{att} and obtain a vector whose feature dimension is $6C$ as the output of the AFF module (see Fig. 5).

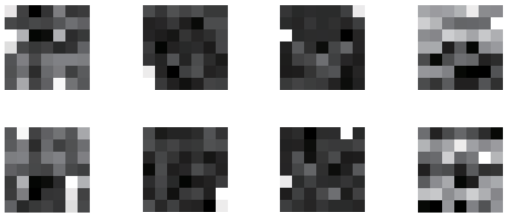


Fig. 6. Outputs of the spatially shuffled samples from the IP dataset for the sixteenth channel. (Top) Original patches. (Bottom) Patches after shuffle.

D. Two-Branch Network Model

The AFF module can be followed by an FC layer with the cross-entropy loss $\mathcal{L}_{\text{ce}}$ for classification

$$\mathcal{L}_{\text{ce}} = -\frac{1}{N}\sum_{i=1}^{N} \boldsymbol{y}_i^T \log(\hat{\boldsymbol{y}}_i) \tag{13}$$

where N is the number of training samples, $\boldsymbol{y}_i$ is a one-hot label vector of the ith sample, and $\hat{\boldsymbol{y}}_i$ is the predicted label distribution. However, in practice, only a limited labeled sample set is available due to the difficulties of manually labeling [59]. In this context, the trained deep model is prone to overfitting and leads to poor generalization capability of classifying unseen samples with unknown spatial distributions.

To mitigate this problem, we incorporate an auxiliary branch into our model by taking into account the impacts of varying spatial distributions. This auxiliary branch and the one introduced before share the same network structure and weights but differ in two aspects. First, the input HSI cube in the auxiliary branch is rearranged with random spatial shuffle (see Fig. 6) before being fed into the subsequent network. Second, a cosine loss $\mathcal{L}_{\cos}$ is explored after the FC layer in the auxiliary branch

$$\begin{aligned}\mathcal{L}_{\cos} &= \frac{1}{N}\sum_{i=1}^{N} 1 - \cos(\boldsymbol{x}_{i,1}, \boldsymbol{x}_{i,2}) \\ &= \frac{1}{N}\sum_{i=1}^{N} 1 - \frac{\boldsymbol{x}_{i,1}\cdot\boldsymbol{x}_{i,2}}{\|\boldsymbol{x}_{i,1}\|\|\boldsymbol{x}_{i,2}\|}\end{aligned} \tag{14}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:14:31 UTC from IEEE Xplore. Restrictions apply.

where $\boldsymbol{x}_{i,1}$ and $\boldsymbol{x}_{i,2}$ are the outputs of FC layers in the two branches for the ith sample (input cube) and $\cdot$ denotes the dot product. This loss ensures feature consistency learning, i.e., the learned features are close enough even if the input cube has large spatial variations.

Finally, we train the whole two-branch network model based on the following unified object function:

$$\mathcal{L} = \mathcal{L}_{\text{ce}} + \lambda \mathcal{L}_{\cos} \tag{15}$$

where λ is a parameter used to balance the two loss terms. It is noted that, after the whole network model is trained, only the first branch is used to predict the classes of new samples.

The proposed model mitigates the overfitting problem via parameter sharing and improves the generalization capability by introducing the spatial shuffle mechanism. By treating the cosine loss as a regularization term, the adopted spatial shuffle mechanism enables our model to perform joint classification and feature consistency learning and, thus, differs from the conventional data augmentation operations.

III. Experiments and Performance Evaluation

A. Datasets and Experimental Setting

We evaluate MSLAN for HSI classification on four publicly available datasets that are described as follows.

1) Indian Pines (IP): The IP dataset was gathered by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor [60] in Northwestern Indiana. It has 224 spectral bands in the wavelength range of 400–2500 nm. We remove the bands covering the region of water absorption, and the number of spectral bands is reduced to 200. The image has 16 classes of vegetation, and the image size is 145 × 145 with a spatial resolution of 3.7 m.

2) University of Pavia (UP): The UP dataset was captured by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor [61] over the UP, Northern Italy. It contains 610 × 340 pixels, and the geometric resolution is 1.3 m. This dataset has nine classes of land covers and 103 spectral bands in the wavelength range of 430–860 nm.

3) University of Houston (UH): This dataset was acquired over the UH campus and the neighboring urban area. The HSI has 144 spectral bands with wavelengths ranging from 380 to 1050 nm, and the spatial size is 349 × 1905 with a spatial resolution of 2.5 m. There are 15 land-cover classes, including grass, trees, water, soil, and some city buildings [62].

4) WHU-Hi-HanChuan (HC): This is a UAV-borne hyperspectral and high spatial resolution dataset acquired in Hanchuan, Hubei province, China [63]. The size of imagery is 1217 × 303 pixels with 274 bands from 400 to 1000 nm, and the spatial resolution is about 0.109 m. This dataset has 16 land-cover classes containing seven crop species used for precise crop classification and HSI classification.

Figs. 7–10 show the false-color images and ground truth for the four datasets. The left-hand side of Tables I–IV shows the specific number of training and test samples. Generally, we set the number of training samples per class as 30 so that we can evaluate different models' generalization capability with limited training samples. For the proposed model, we set the number of principal components after PCA for IP, UP, UH, and HC datasets as 100, 80, 80, and 100, respectively. We set the spatial size of input cubes as 7 × 7 and set $C = 50$ in

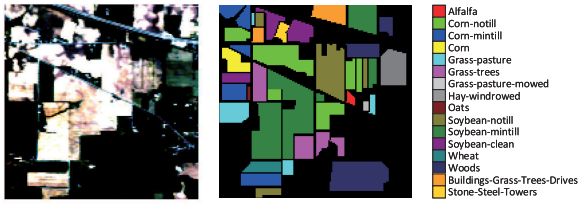


Fig. 7. IP dataset. (a) False-color image. (b) Ground truth. (c) Class names.

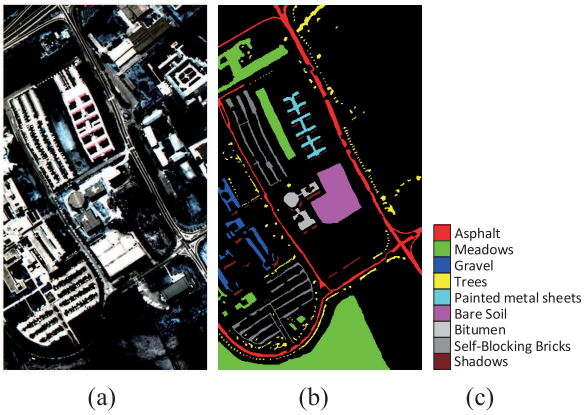


Fig. 8. UP dataset. (a) False-color image. (b) Ground truth. (c) Class names.

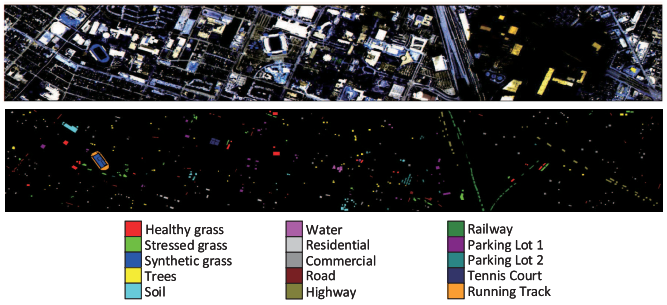


Fig. 9. UH dataset. (Top) False-color image. (Middle) Ground truth. (Bottom) Class names.

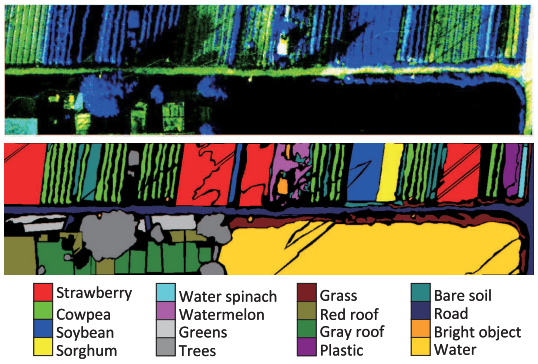


Fig. 10. HC dataset. (Top) False-color image. (Middle) Ground truth. (Bottom) Class names.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:14:31 UTC from IEEE Xplore. Restrictions apply.

TABLE I
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE IP DATASET (THE BEST RESULTS ARE IN BOLD)

Class	Training (#)	Test (#)	LSTM	GRU	SSUN	SSRN	Hybrid-SN	A^2S^2K-ResNet	FuSE-Net	ASSMN	MSLAN-SPA	MSLAN-SPC	MSLAN-	MSLAN
1	30	16	93.75	91.66	100.0	74.24	100.0	70.95	76.74	100.0	100.0	100.0	98.03	100.0
2	30	1398	63.56	59.05	78.18	85.90	71.79	87.50	88.29	82.47	88.66	84.15	92.84	95.90
3	30	800	60.62	58.20	91.12	92.55	87.54	84.38	89.38	93.70	90.07	81.23	90.85	92.03
4	30	207	86.15	79.38	99.19	83.87	95.00	91.25	87.19	98.38	81.11	81.45	96.43	89.19
5	30	453	83.00	82.70	97.05	98.11	94.84	95.36	85.16	93.81	97.80	92.38	96.95	93.93
6	30	700	94.43	91.94	95.71	98.31	94.75	96.81	90.35	98.61	99.47	96.39	99.15	97.67
7	21	7	95.23	90.47	100.0	46.04	100.0	70.64	90.00	100.0	78.33	100.0	54.58	95.83
8	30	448	92.85	90.69	99.85	100.0	100.0	99.92	96.70	100.0	100.0	98.96	100.0	100.0
9	15	5	100.0	93.33	100.0	55.62	100.0	51.85	40.36	100.0	50.92	94.44	57.26	94.44
10	30	942	69.84	74.94	83.71	78.24	86.90	80.65	90.15	89.80	84.94	75.85	84.98	88.22
11	30	2425	53.59	57.37	77.61	94.55	73.67	97.15	95.90	78.92	97.52	87.76	94.41	96.72
12	30	563	63.93	64.06	90.22	83.53	81.70	75.62	80.38	87.50	84.75	74.45	87.19	88.89
13	30	175	96.38	97.71	99.23	93.28	99.42	96.90	88.59	99.42	100.0	93.61	99.62	100.0
14	30	1235	84.34	82.18	92.38	98.98	94.46	98.21	94.42	92.30	99.30	97.01	98.77	98.43
15	30	356	61.79	70.22	99.24	85.80	94.18	89.50	84.35	99.15	81.75	81.69	87.37	91.84
16	30	63	97.88	95.23	100.0	79.82	100.0	90.82	66.89	100.0	89.20	79.31	78.89	87.16
Kappa Coefficient ($\kappa\times100$)			66.02 ±0.78	66.29 ±0.91	85.25 ±0.96	89.04 ±1.49	82.43 ±0.91	89.18 ±0.61	88.09 ±2.08	87.07 ±1.86	91.48 ±0.31	84.74 ±0.45	92.15 ±0.28	**93.95 ±0.21**
Overall Accuracy (%)			69.86 ±0.73	70.11 ±0.79	87.02 ±0.84	90.37 ±1.35	84.48 ±0.88	90.47 ±0.56	89.53 ±1.85	88.63 ±1.65	92.52 ±0.27	86.62 ±0.37	93.13 ±0.24	**94.71 ±0.18**
Average Accuracy (%)			81.08 ±0.73	79.97 ±1.70	93.97 ±0.48	84.30 ±4.05	92.14 ±0.16	86.09 ±2.7	84.05 ±0.71	**94.63 ±0.74**	88.98 ±1.24	88.67 ±0.39	88.58 ±0.86	94.39 ±0.90

Figs. 3–5. The number of neurons in two FC layers in (8) and (9) is 16 and 50, respectively. The hyperparameter λ in (15) is set as 0.4. We use Adam [64] as the optimizer by setting the optimum learning rate as 0.001. The batch size of input data during training is 16. We evaluate the HSI classification performance in terms of the Kappa coefficient (κ), overall accuracy (OA), and average accuracy (AA) by performing experiments on a workstation with an Intel Xeon W-2133 3.60-GHz processor and two NVIDIA GEFORCE RTX 2080 Ti 11-GB GPUs.

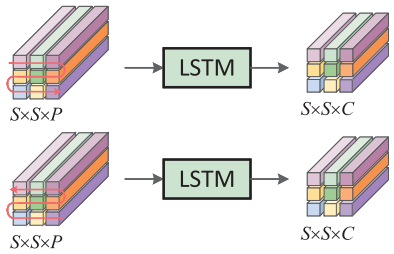


Fig. 11. Two S-shaped spatial directions in MSLAN-.

B. Comparison to State-of-the-Art Methods

We compare the state-of-the-art methods, including LSTM [43], GRU [43], SSUN [48], SSRN [23], HybridSN [22], A^2S^2K-ResNet [38], FuSENet [39], and ASSMN [55]. We implement these methods based on the codes published by the authors. Among them, LSTM and GRU only use spectral vectors as input. SSRN, HybridSN, A^2S^2K-ResNet, and FuSENet are CNN-based methods that use HSI cubes (the spatial size is the same as ours) as input to extract spectral–spatial features, as introduced in Section I. SSUN uses a band grouping-based LSTM and a multiscale CNN to extract spectral and spatial features, respectively. ASSMN uses the band grouping-based LSTM and spatial grouping-based ConvLSTM to extract multiscale spectral and spatial features, respectively. The spatial size of input patches for SSUN and ASSMN is 28 × 28 and 27 × 27, respectively. We also compare three different versions of MSLAN: MSLAN-SPC, MSLAN-SPA, and MSLAN-. MSLAN-SPC uses two spectral directions (forward and backward), while the MSLAN-SPA uses four spatial directions (rightward, leftward, upward, and downward). MSLAN- uses two spatial directions (with S-shape in Fig. 11) and two spectral directions (forward and backward).

Tables I–IV show the classification performance of different methods on the IP, UP, UH, and HC datasets, respectively. Both LSTM and GRU only use spectral information to represent single pixel and yield poor classification results on all four datasets. SSUN integrates the spatial and spectral features into a unified framework and achieves better performance than LSTM and GRU. ASSMN explores regular and ConvLSTMs to learn spectral and spatial features and outperforms SSUN on the IP and HC datasets. However, its performance is worse than that of SSUN on the other two datasets. For CNN-based methods, SSRN uses the well-designed spectral and spatial residual blocks and obtains better results than HybridSN in terms of κ and OA on IP, UP, and UH. However, HybridSN performs better than SSRN in terms of all measures on HC. Particularly, HybridSN achieves the highest AA value among all the compared methods on HC. For attention-based CNN methods, A^2S^2K-ResNet outperforms FuSENet on all four datasets except HC.

For the different versions of MSLAN, MSLAN-SPC that captures two-directional spectral dependencies generally outperforms LSTM and GRU by a large margin. MSLAN-SPA that captures four-directional spatial dependencies shows better performance than MSLAN-SPC. Moreover, MSLAN-SPA outperforms all the other methods except MSLAN in terms of κ and OA on the UH and HC datasets. The intrinsic rich spatial dimensions can contribute to more discriminative features for HSI classification. By using two S-shaped spatial directions

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:14:31 UTC from IEEE Xplore. Restrictions apply.

TABLE II
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE UP DATASET (THE BEST RESULTS ARE IN BOLD)

Class	Training (#)	Test (#)	LSTM	GRU	SSUN	SSRN	Hybrid-SN	A^2S^2K-ResNet	FuSE-Net	ASSMN	MSLAN-SPA	MSLAN-SPC	MSLAN-	MSLAN
1	30	6601	64.78	71.08	88.52	99.59	78.54	97.49	94.86	87.93	97.53	83.65	95.63	96.05
2	30	18619	61.45	65.77	88.93	99.45	96.22	97.83	99.62	83.73	98.62	95.85	97.93	98.81
3	30	2069	71.52	72.12	86.27	72.57	95.64	83.55	79.65	81.02	79.66	68.47	84.28	87.73
4	30	3034	94.25	92.51	97.86	87.43	90.82	93.76	75.66	94.96	89.05	71.60	73.39	95.19
5	30	1315	99.33	99.31	99.18	100.0	99.74	99.74	98.86	99.94	99.86	99.72	99.94	99.97
6	30	4999	62.36	69.19	85.51	92.08	98.69	95.52	98.36	80.69	93.63	77.17	89.72	96.97
7	30	1300	90.94	89.91	92.58	88.52	99.89	87.06	92.15	96.40	92.29	76.00	96.11	94.85
8	30	3652	71.44	81.13	92.13	92.15	88.50	84.24	92.76	91.72	78.33	74.84	72.43	88.62
9	30	917	99.23	99.70	99.70	99.56	94.14	97.88	91.28	100.0	90.97	50.44	92.95	93.59
Kappa Coefficient ($\kappa\times100$)			60.87 ±1.22	66.11 ±1.04	86.81 ±0.93	92.70 ±2.96	90.60 ±0.78	92.93 ±1.14	92.37 ±1.52	82.65 ±2.57	91.57 ±0.02	78.36 ±2.82	87.79 ±1.61	**95.02 ±0.48**
Overall Accuracy (%)			68.65 ±1.11	73.05 ±1.26	89.91 ±0.72	94.40 ±2.32	92.86 ±0.56	94.67 ±0.88	94.17 ±1.20	86.62 ±2.04	93.58 ±0.03	83.36 ±2.23	90.63 ±1.28	**96.25 ±0.35**
Average Accuracy (%)			79.48 ±0.63	82.30 ±1.80	92.30 ±0.79	92.37 ±2.00	93.58 ±0.65	93.01 ±1.73	91.47 ±2.04	90.71 ±1.58	91.11 ±0.68	77.53 ±2.75	89.15 ±1.29	**94.64 ±0.29**

TABLE III
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE UH DATASET (THE BEST RESULTS ARE IN BOLD)

Class	Training (#)	Test (#)	LSTM	GRU	SSUN	SSRN	Hybrid-SN	A^2S^2K-ResNet	FuSE-Net	ASSMN	MSLAN-SPA	MSLAN-SPC	MSLAN-	MSLAN
1	30	1221	93.88	94.88	95.82	89.92	93.09	93.34	92.61	96.88	93.31	89.01	92.82	99.23
2	30	1224	94.30	92.83	98.50	98.25	93.02	98.38	96.38	97.35	98.42	93.28	95.57	98.87
3	30	667	98.80	99.00	99.40	100.0	98.75	100.0	99.90	99.30	100.0	100.0	100.0	99.55
4	30	1214	94.75	92.83	95.07	97.17	94.66	98.53	89.81	94.03	97.94	89.72	95.75	95.85
5	30	1212	96.14	95.97	97.49	98.27	100.0	95.31	99.75	98.59	94.94	99.11	95.79	96.92
6	30	295	91.74	95.70	97.51	99.88	98.30	93.01	92.29	93.77	98.59	98.29	97.35	99.65
7	30	1238	76.22	75.11	90.25	91.86	77.97	92.39	87.13	92.35	93.43	84.78	91.71	94.19
8	30	1214	62.02	55.13	76.38	96.73	84.04	94.50	97.88	72.15	96.68	92.08	96.03	98.82
9	30	1222	74.02	69.36	86.73	90.66	74.13	88.12	88.61	89.95	94.87	85.31	90.55	94.50
10	30	1197	81.64	80.75	89.30	89.44	98.51	89.64	85.63	93.48	89.42	78.81	83.89	88.69
11	30	1205	70.37	74.37	93.55	96.17	96.43	92.32	95.39	89.78	93.77	92.89	92.67	94.75
12	30	1203	58.65	65.33	84.89	88.59	95.53	92.18	91.73	79.24	88.40	86.73	90.24	92.37
13	30	439	48.21	51.02	86.70	80.29	93.69	88.55	92.92	77.06	84.25	85.46	80.41	89.60
14	30	398	98.90	98.99	98.65	94.96	99.74	98.40	93.52	99.91	97.98	94.74	94.68	97.03
15	30	630	98.56	98.67	99.09	97.88	99.46	98.53	93.43	99.04	99.01	98.81	99.31	97.35
Kappa Coefficient ($\kappa\times100$)			80.16 ±0.45	79.87 ±0.52	91.12 ±0.96	92.96 ±0.75	91.27 ±0.03	93.17 ±1.20	91.86 ±1.41	90.38 ±0.24	93.79 ±0.34	89.21 ±0.67	91.93 ±1.10	**95.07 ±0.28**
Overall Accuracy (%)			81.65 ±0.41	81.37 ±0.48	91.78 ±0.89	93.48 ±0.69	91.92 ±0.02	93.69 ±1.11	92.47 ±1.31	91.10 ±0.23	94.25 ±0.32	90.02 ±0.61	92.54 ±1.02	**95.45 ±0.26**
Average Accuracy (%)			82.55 ±0.14	82.66 ±0.18	92.62 ±0.45	94.01 ±0.65	93.15 ±0.09	94.21 ±0.82	93.13 ±0.86	91.52 ±0.28	94.73 ±0.53	91.27 ±0.13	93.12 ±0.84	**95.83 ±0.45**

together with MSLAN-SPC, MSLAN- obtains better results than MSLAN-SPC on the four datasets, but it performs worse than MSLAN-SPA on UP and UH. In contrast, by integrating MSLAN-SPA with MSLAN-SPC, MSLAN achieves almost the best performance on all the four datasets with very limited training samples (only its AA value is not the highest one on IP and HC), demonstrating that the six-directional spatial–spectral features are complementary for HSI classification.

Figs. 12–15 present the classification maps of different methods on the IP, UP, UH, and HC datasets, respectively. By comparing the ground truth in Figs. 7–10, one can see that LSTM wrongly classifies many pixels relying only on spectral features and produces a noise-like classification map. ASSMN reduces such results by further extracting spatial features. SSRN and A^2S^2K-ResNet also give many noisy results for precise crop classification on the HC dataset. HybridSN and FuSENet tend to have smoothed classification maps that can lead to misclassification of some edge or regional pixels. Compared to the above methods, the proposed MSLAN can relatively well retain the structure information while achieving high classification accuracies.

C. Analysis of MSLAN

1) Impact of Spatial Size: The spatial size of input cubes is related to the extraction of spectral–spatial features in MSLAN, which affects the final classification performance. As shown in Fig. 16, the values of OA, AA, and κ rapidly increase when the spatial size changes from 3×3 to 7×7. After that, the performance generally tends to drop. The use of size 7×7 can consistently obtain the best performance on all three datasets. In Fig. 17, we further illustrate the classification maps of the IP dataset by continuously varying the spatial size from 3×3 to 13×13. We can see that small spatial sizes easily lead to noise-like classification results, while large ones tend to yield oversmoothed classification maps. Based on the above quantitative and qualitative analyses, we have chosen the spatial size 7×7 for all datasets by default.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:14:31 UTC from IEEE Xplore. Restrictions apply.

TABLE IV

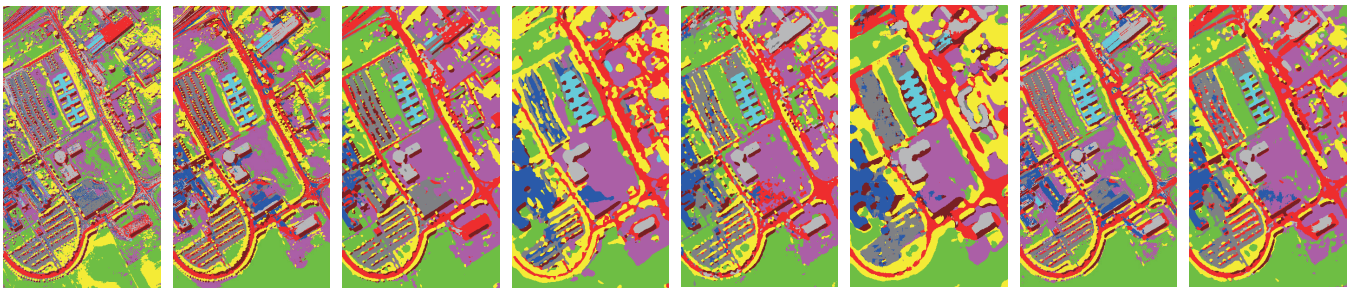
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE HC DATASET (THE BEST RESULTS ARE IN BOLD)

Class	Training (#)	Test (#)	LSTM	GRU	SSUN	SSRN	Hybrid-SN	A^2S^2K-ResNet	FuSE-Net	ASSMN	MSLAN-SPA	MSLAN-SPC	MSLAN-	MSLAN
1	30	44705	75.86	75.17	80.92	93.29	84.13	94.13	97.18	78.59	95.57	85.05	97.05	96.25
2	30	22723	42.88	26.94	67.51	94.99	80.63	93.25	85.50	89.35	94.11	79.82	94.82	94.11
3	30	10257	71.65	60.10	82.40	63.61	82.38	83.24	82.97	82.54	81.05	67.04	88.19	88.04
4	30	5323	93.23	92.04	93.56	85.63	99.82	93.31	99.24	95.46	86.40	91.94	98.71	93.99
5	30	1170	74.29	76.94	97.20	59.23	99.17	50.26	69.72	99.71	42.18	20.82	38.91	45.92
6	30	4503	29.33	35.99	69.60	29.43	80.62	52.14	60.94	62.85	50.10	36.46	52.20	64.05
7	30	5873	85.97	81.09	85.43	49.25	94.24	58.60	78.67	90.32	67.37	65.55	65.73	71.11
8	30	17948	56.96	54.65	73.94	88.38	72.46	87.99	89.27	71.30	89.65	74.46	88.14	94.16
9	30	9439	37.99	43.14	73.06	83.33	73.84	73.35	81.31	75.12	73.21	54.26	74.92	75.17
10	30	10486	87.91	88.09	91.69	83.10	95.37	93.62	96.18	88.34	99.27	87.21	98.95	99.60
11	30	16881	85.15	85.59	85.45	70.50	93.44	67.34	88.95	79.17	81.29	77.98	85.35	88.38
12	30	3649	56.17	58.95	78.00	57.67	84.00	48.52	38.83	94.83	55.89	41.96	51.01	62.73
13	30	9086	48.62	46.36	68.98	40.92	71.78	51.42	58.89	70.35	65.97	42.97	57.97	64.03
14	30	18530	56.22	56.70	81.09	82.44	76.13	87.45	91.12	85.95	91.38	71.94	84.72	93.08
15	30	1106	82.01	82.21	91.13	56.49	88.81	72.94	43.67	95.77	59.74	32.40	60.89	90.61
16	30	75371	91.88	92.35	90.86	98.98	97.68	99.55	99.98	94.97	99.96	98.77	99.82	99.74
Kappa Coefficient ($\kappa\times100$)			68.91 ±1.34	66.85 ±0.70	79.79 ±1.98	74.63 ±0.37	85.09 ±0.31	81.83 ±2.25	84.64 ±1.58	82.94 ±0.89	86.31 ±0.71	74.77 ±0.87	86.29 ±0.65	**89.35 ±0.41**
Overall Accuracy (%)			73.00 ±1.22	71.20 ±0.59	82.50 ±1.78	78.01 ±0.32	87.17 ±0.29	84.27 ±2.03	86.77 ±1.36	85.28 ±0.79	88.21 ±0.62	78.31 ±0.69	88.20 ±0.57	**90.86 ±0.36**
Average Accuracy (%)			67.26 ±0.70	66.02 ±0.84	81.92 ±1.14	71.08 ±0.19	**85.90 ±1.24**	75.45 ±0.43	78.90 ±0.19	84.67 ±0.91	77.07 ±1.26	64.29 ±1.14	77.34 ±1.50	82.56 ±1.18



(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 12. Classification maps of the IP dataset. (a) LSTM. (b) SSUN. (c) SSRN. (d) HybridSN. (e) A^2S^2K-ResNet. (f) FuSENet. (g) ASSMN. (h) MSLAN.



(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 13. Classification maps of the UP dataset. (a) LSTM. (b) SSUN. (c) SSRN. (d) HybridSN. (e) A^2S^2K-ResNet. (f) FuSENet. (g) ASSMN. (h) MSLAN.

2) Impact of Cosine Loss: In our MSLAN, a cosine loss is used for feature consistency learning between two branches. Alternative losses based on $L1$- and $L2$-norms can be used, which are defined as follows:

$$\mathcal{L}_{L1} = \frac{1}{N}\sum_{i=1}^{N}\left\|\boldsymbol{x}_{i,1}-\boldsymbol{x}_{i,2}\right\|_1 \tag{16}$$

$$\mathcal{L}_{L2} = \frac{1}{N}\sum_{i=1}^{N}\left\|\boldsymbol{x}_{i,1}-\boldsymbol{x}_{i,2}\right\|_2^2. \tag{17}$$

As shown in Table V, $\mathcal{L}_{\cos}$ performs the best on all the three datasets. This is because the cosine loss is more concerned about the direction information of two vectors compared to the other two losses. For feature consistency learning, the compared features in (14) can be the outputs of two AFFs

TABLE V

IMPACT OF DIFFERENT LOSSES ON THE CLASSIFICATION RESULTS

	IP			UP			UH		
	$\kappa\times100$	OA	AA	$\kappa\times100$	OA	AA	$\kappa\times100$	OA	AA
$\mathcal{L}_{L1}$	92.51 ±0.21	93.43 ±0.18	90.00 ±0.17	92.97 ±0.51	94.66 ±0.39	92.21 ±0.73	94.71 ±0.70	95.11 ±0.65	94.75 ±0.89
$\mathcal{L}_{L2}$	92.45 ±1.38	93.38 ±1.23	89.92 ±1.86	91.06 ±0.75	93.16 ±0.58	91.73 ±0.86	94.58 ±0.24	94.99 ±0.22	95.24 ±0.15
$\mathcal{L}_{cos}$	**93.95 ±0.21**	**94.71 ±0.18**	**94.39 ±0.90**	**95.02 ±0.48**	**96.25 ±0.35**	**94.64 ±0.29**	**95.07 ±0.28**	**95.45 ±0.26**	**95.83 ±0.45**

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:14:31 UTC from IEEE Xplore. Restrictions apply.

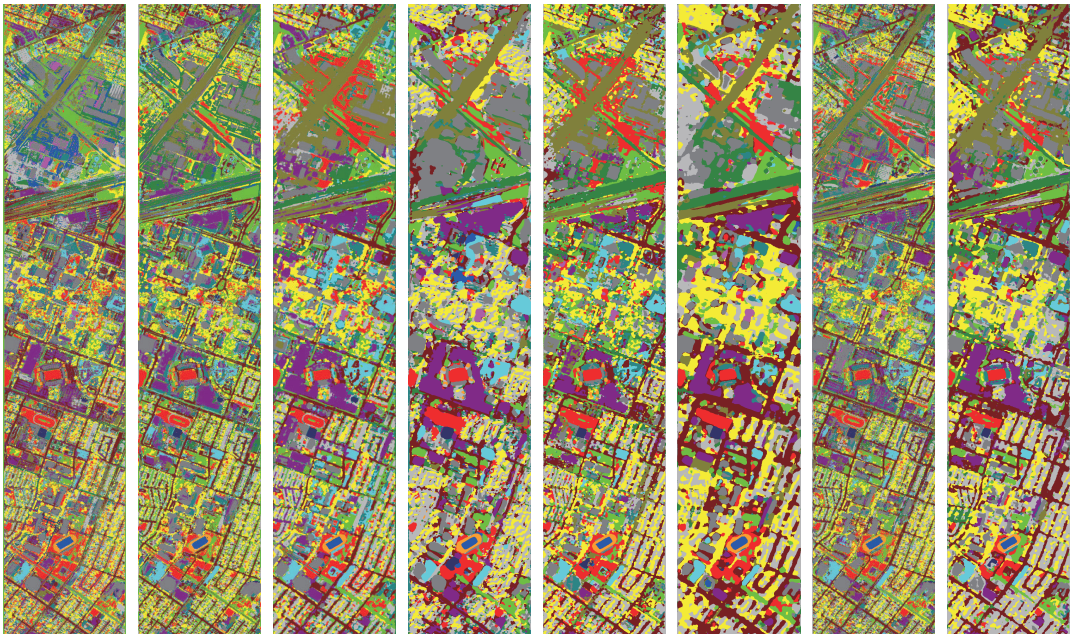


Fig. 14. Classification maps of the UH dataset. (a) LSTM. (b) SSUN. (c) SSRN. (d) HybridSN. (e) A^{2}S^{2}K-ResNet. (f) FuSENet. (g) ASSMN. (h) MSLAN.

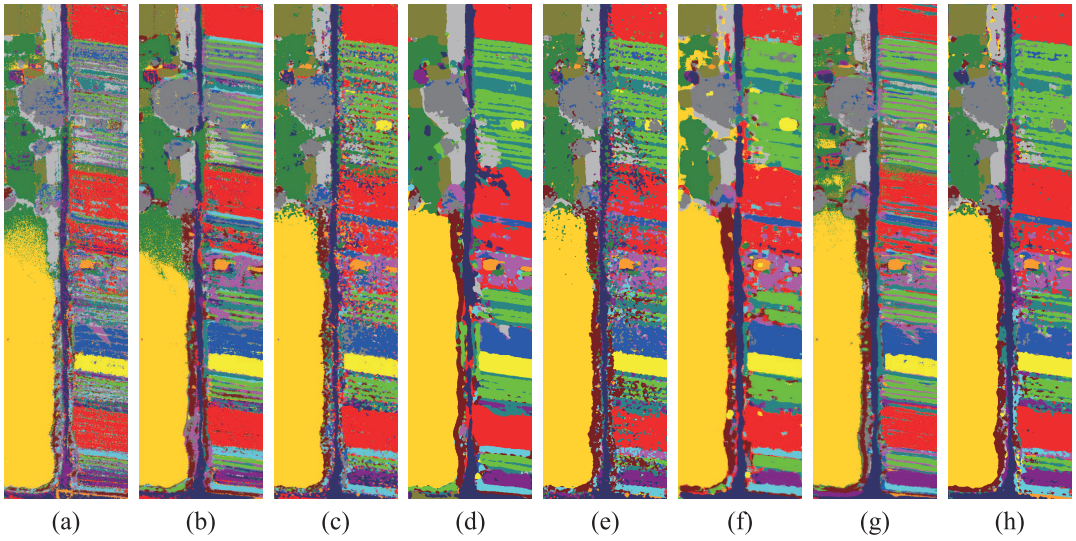


Fig. 15. Classification maps of the HC dataset. (a) LSTM. (b) SSUN. (c) SSRN. (d) HybridSN. (e) A^{2}S^{2}K-ResNet. (f) FuSENet. (g) ASSMN. (h) MSLAN.

(denoted as CONV_300 with feature length 300) or the outputs of two FCs shown in Fig. 1. Also, we add a new FC between AFF and the current FC in the first branch. The new FC shares weights with the FC in the second branch, and their outputs are denoted as FC_150 (FC_300) with feature length 150 (300). From Table VI, we find that the best results are obtained based on the current FCs. How to theoretically select the best compared features remains an open question.

3) Impact of Different Components: The proposed MSLAN includes four major components, i.e., the PCA operation, the multidirectional LSTM-based module, the AFF module, and the shuffle-based auxiliary branch. To conduct an ablation study, we define a baseline method, i.e., the input cubes (without using PCA) are directly fed into the multidirectional LSTM-based module, followed by an FC layer with a cross-entropy loss for classification. As shown in Table VII, the use of each module alone cannot achieve the best performance. However, the addition of one more module can almost

TABLE VI

IMPACT OF FEATURE CONSISTENCY LEARNING ON THE CLASSIFICATION RESULTS BY COMPARING DIFFERENT FEATURES

	IP			UP			UH		
	$\kappa\times100$	OA	AA	$\kappa\times100$	OA	AA	$\kappa\times100$	OA	AA
FC_150	89.88 ±0.61	91.11 ±0.55	91.81 ±0.37	92.00 ±0.48	93.91 ±0.38	91.54 ±0.35	93.92 ±0.31	94.38 ±0.28	94.69 ±0.35
FC_300	91.04 ±0.36	92.13 ±0.31	91.53 ±0.91	90.75 ±0.32	92.94 ±0.25	90.75 ±0.02	93.18 ±0.17	93.69 ±0.16	94.32 ±0.12
CONV_300	92.95 ±0.97	93.82 ±0.85	91.43 ±0.77	90.32 ±0.47	92.60 ±0.36	90.25 ±0.78	94.60 ±0.19	95.01 ±0.17	95.31 ±0.37
FC	**93.95 ±0.21**	**94.71 ±0.18**	**94.39 ±0.90**	**95.02 ±0.48**	**96.25 ±0.35**	**94.64 ±0.29**	**95.07 ±0.28**	**95.45 ±0.26**	**95.83 ±0.45**

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:14:31 UTC from IEEE Xplore. Restrictions apply.

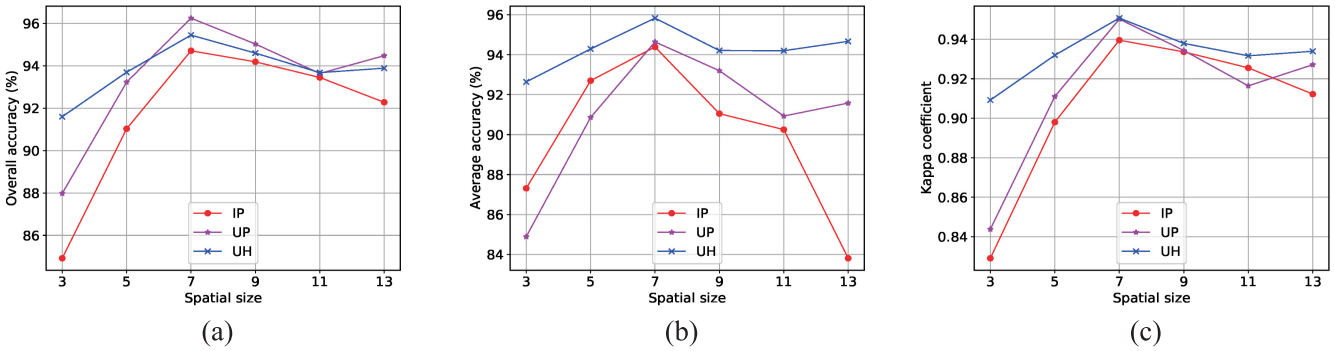


Fig. 16. Impact of the spatial size on the classification performance. (a) OA. (b) AA. (c) Kappa coefficient.

TABLE VII
IMPACT OF DIFFERENT COMPONENTS ON THE CLASSIFICATION RESULTS

	Components					IP			UP			UH		
	PCA	AFF	SE	Shuffle	Aug.	$\kappa\times100$	OA	AA	$\kappa\times100$	OA	AA	$\kappa\times100$	OA	AA
Baseline						81.12±0.27	83.37±0.23	81.39±0.98	85.33±1.79	88.84±1.45	88.26±1.36	90.33±0.24	91.06±0.23	92.01±0.07
+ PCA	✓					85.08±0.36	86.87±0.31	85.21±0.73	88.84±0.39	91.51±0.30	88.85±0.86	91.30±0.82	91.95±0.76	92.89±0.38
+ AFF		✓				81.34±1.12	83.61±0.99	78.50±0.94	87.62±0.34	90.61±0.20	88.62±0.40	89.92±0.10	90.68±0.09	91.53±0.10
+ PCA + AFF	✓	✓				89.31±0.55	90.62±0.51	89.34±0.84	88.71±0.15	91.39±0.10	88.62±0.54	92.90±0.13	93.43±0.12	94.23±0.23
+ PCA + SE + Shuffle	✓		✓	✓		91.19±0.73	92.28±0.64	91.11±0.76	91.02±0.72	93.14±0.57	90.84±0.97	93.40±0.57	93.90±0.52	94.34±0.37
+ PCA + AFF + Aug.	✓	✓			✓	91.26±0.48	92.34±0.43	87.38±0.52	90.44±0.60	92.70±0.49	90.42±0.50	93.80±0.51	94.27±0.47	94.52±0.31
+ PCA + AFF + Shuffle	✓	✓		✓		**93.95±0.21**	**94.71±0.18**	**94.39±0.90**	**95.02±0.48**	**96.25±0.35**	**94.64±0.29**	**95.07±0.28**	**95.45±0.26**	**95.83±0.45**

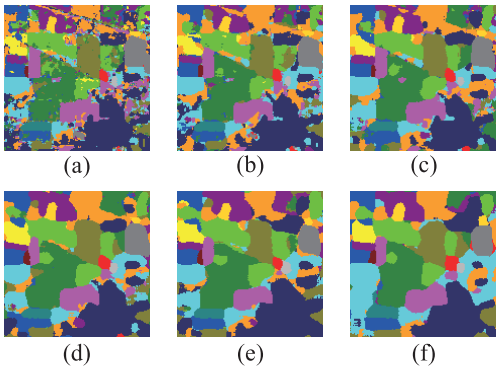


Fig. 17. Classification maps of the IP dataset for MSLAN using different spatial sizes. (a) 3 × 3. (b) 5 × 5. (c) 7 × 7. (d) 9 × 9. (e) 11 × 11. (f) 13 × 13.

TABLE VIII
IMPACT OF DIFFERENT SEQUENCE MODELING SCHEMES ON THE CLASSIFICATION RESULTS

	IP			UP			UH		
	$\kappa\times100$	OA	AA	$\kappa\times100$	OA	AA	$\kappa\times100$	OA	AA
RNNs	90.80 ±0.24	91.92 ±0.22	89.65 ±0.16	92.58 ±0.31	94.38 ±0.24	92.85 ±0.67	94.33 ±0.46	94.76 ±0.43	95.12 ±0.35
GRUs	93.07 ±0.28	93.93 ±0.25	92.77 ±2.46	93.19 ±1.16	94.36 ±0.68	93.69 ±1.34	93.39 ±0.21	93.89 ±0.19	94.45 ±0.20
TCNs	90.87 ±1.23	91.99 ±1.09	93.37 ±0.42	92.69 ±0.88	94.46 ±0.67	92.06 ±0.87	93.57 ±0.50	94.05 ±0.46	94.75 ±0.64
LSTMs	**93.95 ±0.21**	**94.71 ±0.18**	**94.39 ±0.90**	**95.02 ±0.48**	**96.25 ±0.35**	**94.64 ±0.29**	**95.07 ±0.28**	**95.45 ±0.26**	**95.83 ±0.45**

significantly improve the classification accuracy. By combining all the four components, our method achieves the best results on all three datasets. Note that, in this work, we have used the prior knowledge of PCA to remove noise and information redundancy [21], [39], [48], [55], thereby reducing the number of model parameters and improving the training efficiency of our deep network. In future work, we will consider the joint dimensionality reduction and classification in a unified deep learning framework.

Moreover, we replace the AFF module with the channelwise SE block [32]. Table VII shows that the AFF module achieves better classification performance than the SE block. Hence, by adopting groupwise direction attention, AFF is well adapted for our multidirectional LSTM network. We also discard the spatial shuffle branch and use its counterpart based on the conventional data augmentation (written as "Aug."). As shown in Table VII, our spatial shuffle mechanism performs better than the conventional one, indicating the effectiveness of joint classification and feature consistency learning.

4) Impact of Different Sequence Modeling Schemes: We compare four different multidirectional sequence modeling schemes, including three recurrent architectures (RNNs, GRUs, and LSTMs) and one convolutional architecture called the generic temporal convolutional network (TCN) [65] (1-D fully convolutional network architecture with causal convolutions). The results are shown in Table VIII. Overall, GRUs perform better than RNNs and TCNs on the IP and UP datasets, while RNNs perform better than GRUs and TCNs on the UH dataset. Among these four schemes, LSTMs consistently achieve the best results. Hence, in this work, we have selected LSTMs to extract multidirectional spatial–spectral features.

5) Convergence Curves: Fig. 18 plots the curves of cross-entropy loss and cosine loss for training MSLAN with respect to the number of epochs on the three datasets. We can see that the curve of cross-entropy loss flattens out after about ten epochs. For the cosine loss, it converges after about

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:14:31 UTC from IEEE Xplore. Restrictions apply.

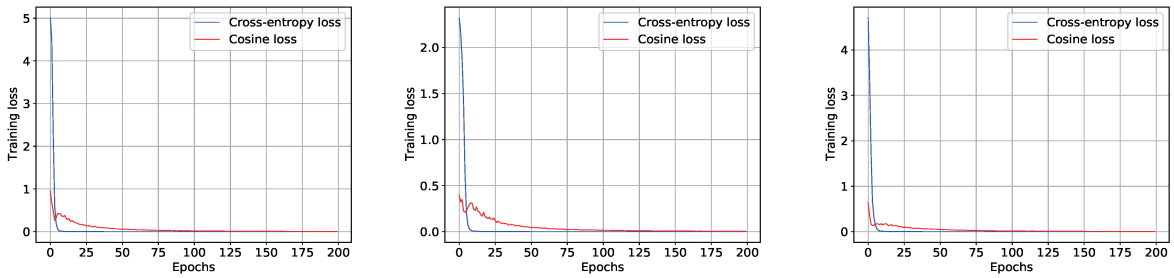


Fig. 18. Curves of training loss on three datasets. (a) IP. (b) UP. (c) UH.

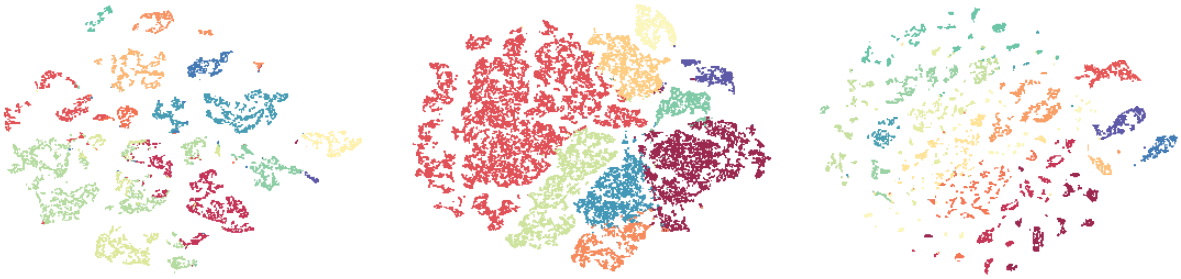


Fig. 19. Visualization of the extracted features via t-SNE for the test samples in three datasets. (a) IP. (b) UP. (c) UH.

TABLE IX

RUNNING TIME (IN SECONDS) AND THE NUMBER OF TRAINABLE PARAMETERS REQUIRED BY DIFFERENT METHODS ON THE IP DATASET

	LSTM	GRU	SSUN	SSRN	Hybrid-SN	A^2S^2K-ResNet	FuSE-Net	ASSMN	MSLAN
Training	1.62	1.56	3.50	6.96	4.83	3.21	9.08	3.81	5.27
Test	0.60	0.59	0.93	2.26	2.13	2.49	1.85	3.14	4.39
Parameters	1.2K	0.9K	1.2M	346K	5.1M	370K	129K	2.3M	206K

125 epochs. Hence, we set the number of training epochs as 200 for all the datasets.

6) Feature Visualization: As shown in Fig. 19, we map the output features of the FC layer in MSLAN to 2-D space via t-distributed stochastic neighbor embedding (t-SNE) [66] (the land-cover labels are shown in different colors). Although we have trained the model using about 30 samples per class, the sample features for each category in all datasets still have good separability. Thus, the proposed model generalizes well to learn discriminative spectral–spatial feature representations with limited training samples.

7) Computational and Storage Costs: Table IX shows the running time and the number of trainable parameters required by different methods on the IP dataset. The first row presents the training time for one epoch, and the second row represents the test time to predict all test samples. We can see that the training speed of MSLAN is moderate among the compared methods, and MSLAN runs faster than SSRN and FuSENet. On the other hand, the test time of MSLAN is relatively long due to the deployment of multiple directional LSTMs. In addition, our model requires a small number of trainable parameters. Only LSTM, GRU, and FuSENet have fewer parameters than MSLAN among the compared methods.

IV. CONCLUSION

In this article, a two-branch MSLAN is proposed for HSI classification. In MSLAN, multidirectional LSTMs are explored to fully capture the spectral–spatial dependencies in six directions. An AFF module is designed to select and integrate these directional features, followed by a fully connected layer with the cross-entropy loss for classification. To account for the impacts of varying spatial distributions, an auxiliary branch is incorporated into our model, which adopts random spatial shuffle and a cosine loss for feature consistency learning. These two branch networks share the same network structure and weights, which are trained in a unified deep learning framework. Experiments on four datasets demonstrate the effectiveness of MSLAN for HSI classification with limited training samples.

REFERENCES

[1] P. Ghamisi *et al.*, "Advances in hyperspectral image and signal processing: A comprehensive overview of the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 37–78, Dec. 2017.

[2] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Apr. 2019.

[3] P. Ghamisi, J. Plaza, Y. Chen, J. Li, and A. J. Plaza, "Advanced spectral classifiers for hyperspectral images: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 1, pp. 8–32, Mar. 2017.

[4] W. Sun and Q. Du, "Hyperspectral band selection: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 118–139, Jun. 2019.

[5] B. Rasti *et al.*, "Feature extraction for hyperspectral imagery: The evolution from shallow to deep: Overview and toolbox," *IEEE Geosci. Remote Sens. Mag.*, vol. 8, no. 4, pp. 60–88, Dec. 2020.

[6] W. Li, F. Feng, H. Li, and Q. Du, "Discriminant analysis-based dimension reduction for hyperspectral image classification: A survey of the most recent advances and an experimental comparison of different techniques," *IEEE Geosci. Remote Sens. Mag.*, vol. 6, no. 1, pp. 15–34, Mar. 2018.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:14:31 UTC from IEEE Xplore. Restrictions apply.

[7] D. Lunga, S. Prasad, M. M. Crawford, and O. Ersoy, "Manifold-learning-based feature extraction for classification of hyperspectral data: A review of advances in manifold learning," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 55–66, Jan. 2014.

[8] L. He, J. Li, C. Liu, and S. Li, "Recent advances on spectral-spatial hyperspectral image classification: An overview and new guidelines," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 3, pp. 1579–1597, Mar. 2017.

[9] W. Li, C. Chen, H. Su, and Q. Du, "Local binary patterns and extreme learning machine for hyperspectral imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 7, pp. 3681–3693, Jul. 2015.

[10] L. Shen and S. Jia, "Three-dimensional Gabor wavelets for pixel-based hyperspectral imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 12, pp. 5039–5046, Dec. 2011.

[11] M. Fauvel, J. A. Benediktsson, J. Chanussot, and J. R. Sveinsson, "Spectral and spatial classification of hyperspectral data using SVMs and morphological profiles," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 11, pp. 3804–3814, Nov. 2008.

[12] W. Li, S. Prasad, and J. E. Fowler, "Hyperspectral image classification using Gaussian mixture models and Markov random fields," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 1, pp. 153–157, Jan. 2014.

[13] X. Kang, S. Li, and J. A. Benediktsson, "Spectral-spatial hyperspectral image classification with edge-preserving filtering," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 5, pp. 2666–2677, May 2014.

[14] S. Jia, B. Deng, J. Zhu, X. Jia, and Q. Li, "Local binary pattern-based hyperspectral image classification with superpixel guidance," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 749–759, Feb. 2018.

[15] R. Hang, Q. Liu, D. Hong, and P. Ghamisi, "Cascaded recurrent neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5384–5394, Aug. 2019.

[16] L. Zhang, L. Zhang, and B. Du, "Deep learning for remote sensing data: A technical tutorial on the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 4, no. 2, pp. 22–40, Jan. 2016.

[17] Y. Chen, Z. Lin, X. Zhao, G. Wang, and Y. Gu, "Deep learning-based classification of hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2094–2107, Jun. 2014.

[18] Y. Chen, X. Zhao, and X. Jia, "Spectral-spatial classification of hyperspectral data based on deep belief network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2381–2392, Jun. 2015.

[19] W. Hu, Y. Huang, L. Wei, F. Zhang, and H. Li, "Deep convolutional neural networks for hyperspectral image classification," *J. Sensors*, vol. 2015, Jul. 2015, Art. no. 258619.

[20] J. Yue, W. Zhao, S. Mao, and H. Liu, "Spectral-spatial classification of hyperspectral images using deep convolutional neural networks," *Remote Sens. Lett.*, vol. 6, no. 6, pp. 468–477, May 2015.

[21] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Jul. 2016.

[22] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3-D–2-D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb. 2020.

[23] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral-spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2018.

[24] W. Li, G. Wu, F. Zhang, and Q. Du, "Hyperspectral image classification using deep pixel-pair features," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 2, pp. 844–853, Feb. 2017.

[25] M. Zhang, W. Li, and Q. Du, "Diverse region-based CNN for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 27, no. 6, pp. 2623–2634, Jun. 2018.

[26] S. Hao, W. Wang, Y. Ye, T. Nie, and L. Bruzzone, "Two-stream deep architecture for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 2349–2361, Apr. 2018.

[27] N. He *et al.*, "Feature extraction with multiscale covariance maps for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 755–769, Feb. 2019.

[28] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2021.

[29] Q. Liu, L. Xiao, J. Yang, and Z. Wei, "Multilevel superpixel structured graph U-Nets for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022.

[30] L. Zhu, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Generative adversarial networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 9, pp. 5046–5063, Sep. 2018.

[31] A. Vaswani *et al.*, "Attention is all you need," in *Proc. Neural Inf. Process. Syst.*, 2017, pp. 1–11.

[32] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.

[33] J. M. Haut, M. E. Paoletti, J. Plaza, A. Plaza, and J. Li, "Visual attention-driven hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 8065–8080, Oct. 2019.

[34] M. Zhu, L. Jiao, F. Liu, S. Yang, and J. Wang, "Residual spectral–spatial attention network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 449–462, Jan. 2021.

[35] H. Sun, X. Zheng, X. Lu, and S. Wu, "Spectral-spatial attention network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3232–3245, May 2020.

[36] R. Hang, Z. Li, Q. Liu, P. Ghamisi, and S. S. Bhattacharyya, "Hyperspectral image classification with attention-aided CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2281–2293, Mar. 2021.

[37] B. Xi *et al.*, "Multi-direction networks with attentional spectral prior for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022.

[38] S. K. Roy, S. Manna, T. Song, and L. Bruzzone, "Attention-based adaptive spectral–spatial kernel ResNet for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7831–7843, Sep. 2021.

[39] S. K. Roy, S. R. Dubey, S. Chatterjee, and B. B. Chaudhuri, "FuSENet: Fused squeeze-and-excitation network for spectral-spatial hyperspectral image classification," *IET Image Process.*, vol. 14, no. 8, pp. 1653–1661, 2020.

[40] Z. Xue, M. Zhang, Y. Liu, and P. Du, "Attention-based second-order pooling network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9600–9615, Nov. 2021.

[41] L. Mou, P. Ghamisi, and X. X. Zhu, "Deep recurrent neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3639–3655, Jul. 2016.

[42] H. Wu and S. Prasad, "Convolutional recurrent neural networks for hyperspectral data classification," *Remote Sens.*, vol. 9, no. 3, p. 298, 2017.

[43] M. E. Paoletti, J. M. Haut, J. Plaza, and A. Plaza, "Scalable recurrent neural network for hyperspectral image classification," *J. Supercomput.*, vol. 76, pp. 8866–8882, Feb. 2020.

[44] X. Shi, Z. Chen, H. Wang, D. Yeung, W. Wong, and W. Woo, "Convolutional LSTM network: A machine learning approach for precipitation nowcasting," in *Proc. Adv. Neural Inf. Process. Syst.*, 2015, pp. 802–810.

[45] Q. Liu, F. Zhou, R. Hang, and X. Yuan, "Bidirectional-convolutional LSTM based spectral-spatial feature learning for hyperspectral image classification," *Remote Sens.*, vol. 9, no. 12, p. 1330, 2017.

[46] X. Yang, Y. Ye, X. Li, R. Y. K. Lau, X. Zhang, and X. Huang, "Hyperspectral image classification with deep learning models," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 9, pp. 5408–5423, Sep. 2018.

[47] W.-S. Hu, H.-C. Li, L. Pan, W. Li, R. Tao, and Q. Du, "Spatial–spectral feature extraction via deep ConvLSTM neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 4237–4250, Jun. 2020.

[48] Y. Xu, L. Zhang, B. Du, and F. Zhang, "Spectral-spatial unified networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 5893–5909, Oct. 2018.

[49] X. Zhang, Y. Sun, K. Jiang, C. Li, L. Jiao, and H. Zhou, "Spatial sequential recurrent neural network for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 11, pp. 4141–4155, Nov. 2018.

[50] C. Shi and C.-M. Pun, "Multi-scale hierarchical recurrent neural networks for hyperspectral image classification," *Neurocomputing*, vol. 294, pp. 82–93, Jun. 2018.

[51] A. Graves and J. Schmidhuber, "Offline handwriting recognition with multidimensional recurrent neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2008, pp. 545–552.

[52] Z. Zuo *et al.*, "Learning contextual dependence with convolutional hierarchical recurrent neural networks," *IEEE Trans. Image Process.*, vol. 25, no. 7, pp. 2983–2996, Jul. 2016.

[53] B. Liu, X. Yu, A. Yu, P. Zhang, and G. Wan, "Spectral–spatial classification of hyperspectral imagery based on recurrent neural networks," *Remote Sens. Lett.*, vol. 9, no. 12, pp. 1118–1127, 2018.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:14:31 UTC from IEEE Xplore. Restrictions apply.

[54] F. Zhou, R. Hang, Q. Liu, and X. Yuan, "Hyperspectral image classification using spectral-spatial LSTMs," *Neurocomputing*, vol. 328, pp. 39–47, Feb. 2018.
[55] D. Wang, B. Du, L. Zhang, and Y. Xu, "Adaptive spectral-spatial multiscale contextual feature extraction for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2461–2477, Mar. 2020.
[56] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, 1997.
[57] X. Pan, J. Shi, P. Luo, X. Wang, and X. Tang, "Spatial as deep: Spatial CNN for traffic scene understanding," in *Proc. AAAI Conf. Artif. Intell.*, 2018, pp. 7276–7283.
[58] S. Liu, S. De Mello, J. Gu, G. Zhong, M.-H. Yang, and J. Kautz, "Learning affinity via spatial propagation networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 1–11.
[59] S. Jia, S. Jiang, Z. Lin, N. Li, M. Xu, and S. Yu, "A survey: Deep learning for hyperspectral image classification with few labeled samples," *Neurocomputing*, vol. 448, pp. 179–204, Aug. 2021.
[60] R. O. Green *et al.*, "Imaging spectroscopy and the airborne visible/infrared imaging spectrometer (AVIRIS)," *Remote Sens. Environ.*, vol. 65, no. 3, pp. 227–248, Sep. 1998.
[61] B. Kunkel, F. Blechinger, R. Lutz, R. Doerffer, H. Van der Piepen, and M. Schroder, "ROSIS (Reflective optics system imaging spectrometer)—A candidate instrument for polar platform missions," *Proc. SPIE*, vol. 868, pp. 134–141, Apr. 1988.
[62] C. Debes *et al.*, "Hyperspectral and LiDAR data fusion: Outcome of the 2013 GRSS data fusion contest," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2405–2418, Jun. 2014.
[63] Y. Zhong, X. Hu, C. Luo, X. Wang, J. Zhao, and L. Zhang, "WHU-Hi: UAV-borne hyperspectral with high spatial resolution (H^2) benchmark datasets and classifier for precise crop identification based on deep convolutional neural network with CRF," *Remote Sens. Environ.*, vol. 250, Dec. 2020, Art. no. 112012.
[64] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.
[65] S. Bai, J. Zico Kolter, and V. Koltun, "An empirical evaluation of generic convolutional and recurrent networks for sequence modeling," 2018, *arXiv:1803.01271*.
[66] L. Van der Maaten and G. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, no. 11, pp. 2579–2605, 2008.



Tiecheng Song (Member, IEEE) received the Ph.D. degree in signal and information processing from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2015.

From October 2015 to April 2016, he was a Visiting Student with the Multimedia Laboratory, Nanyang Technological University, Singapore. He is currently an Associate Professor with the School of Communication and Information Engineering, Chongqing University of Posts and Telecommunications (CQUPT), Chongqing, China. His research interests include image processing and computer vision.



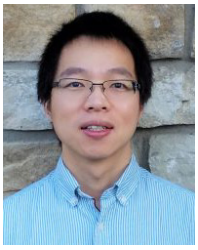
Yuanlin Wang received the B.S. degree in electronic and information engineering from the Chongqing University of Posts and Telecommunications, Chongqing, China, in 2019, where he is currently pursuing the M.S. degree with the School of Communication and Information Engineering.

His research interests include remote image processing and deep learning.



Chenqiang Gao received the Ph.D. degree in control science and engineering from the Huazhong University of Science and Technology, Wuhan, China, in 2009.

In August 2009, he joined the School of Communications and Information Engineering, Chongqing University of Posts and Telecommunications (CQUPT), Chongqing, China. In September 2012, he joined the Informedia Group, School of Computer Science, Carnegie Mellon University, Pittsburgh, PA, USA, where he was a Visiting Scholar on multimedia event detection (MED) and surveillance event detection (SED). In April 2013, he became a Post-Doctoral Fellow at Carnegie Mellon University and continued work on MED and SED until March 2014, when he returned to CQUPT. He is currently a Professor with CQUPT. His research interests include image processing, infrared target detection, action recognition, and event detection.



Haonan Chen (Senior Member, IEEE) received the Ph.D. degree in electrical engineering from Colorado State University (CSU), Fort Collins, CO, USA, in 2017.

He has worked at the Cooperative Institute for Research in the Atmosphere (CIRA), Fort Collins, CO, USA, and the NOAA Physical Sciences Laboratory, Boulder, CO, USA, from 2012 to 2020, first as a Research Student, then a National Research Council Research Associate, and a Radar, Satellite, and Precipitation Research Scientist. He is currently an Assistant Professor with CSU. His research interests lie primarily in the areas of remote sensing and multidisciplinary data sciences, including multifrequency radar and satellite observations of clouds and precipitation, polarimetric radar systems and networking, hydrometeorology data analytics, and deep learning.

Dr. Chen also serves as an Associate Editor for the *Journal of Atmospheric and Oceanic Technology*, *URSI Radio Science Bulletin*, and the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING.



Jun Li (Fellow, IEEE) received the B.S. degree in geographic information systems from Hunan Normal University, Changsha, China, in 2004, the M.E. degree in remote sensing from Peking University, Beijing, China, in 2007, and the Ph.D. degree in electrical engineering from the Instituto de Telecomunicações, Instituto Superior Técnico (IST), Universidade Técnica de Lisboa, Lisbon, Portugal, in 2011.

She is currently a Full Professor with the China University of Geosciences, Wuhan, China, where she founded her own research group, Hyperspectral Calibration and Learning (HCL), in 2013. Since then, she has obtained several prestigious funding grants at the national and international levels. She has published more than 160 journal citation report (JCR) papers, 60 international conference papers, and a book chapter.

Dr. Li has been serving as the Editor-in-Chief of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING since 2021.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:14:31 UTC from IEEE Xplore. Restrictions apply.